# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau.

## An Interactive Tableau Dashboard for Cosmetics Market Analysis and Business Intelligence


**Team ID:**

**Team Members:**

Abdul Hafeez Shaik (Team Lead)

Gollapalli Gayathri

Sana Grace

Avanigadda Dola Chaturya

Revathi Gorupina


## PROJECT FINAL REPORT:

**Date:** 13 July 2026

# 1. INTRODUCTION

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## 1.1 Project Overview

The cosmetics industry is rapidly growing due to changing consumer preferences, product innovation, and increasing demand for personalized skincare and beauty products. Businesses need data-driven insights to understand market trends, evaluate product performance, and improve customer satisfaction.

This project, **"Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,"** analyzes a cosmetics dataset containing information such as brands, product categories, prices, rankings, and skin-type suitability. After cleaning and preparing the data, an interactive Tableau dashboard was developed to visualize key performance indicators and business trends.

The dashboard enables users to compare brand performance, analyze pricing patterns, study product categories, and evaluate products suitable for different skin types. These visual insights support better business decisions in marketing, product development, and customer targeting.

## 1.2 Purpose

The purpose of this project is to transform cosmetic product data into meaningful business insights using Tableau. The project aims to:

- Analyze brand and product category performance.
- Compare pricing trends across cosmetic brands.
- Evaluate product rankings and customer preferences.
- Study product suitability for different skin types.
- Provide interactive dashboards for data-driven decision-making.
- Support businesses in improving marketing and product strategies.

Overall, the project demonstrates how Tableau can convert raw cosmetic data into interactive visualizations that help stakeholders understand market trends and make informed business decisions.

# 2. IDEATION PHASE

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## 2.1 Problem Statement

### Customer Problem Statement Template

A clear problem statement ensures that the project focuses on solving real business challenges faced by stakeholders in the cosmetics industry. Cosmetic companies, retailers, marketing teams, and business analysts often struggle to understand consumer preferences, product performance, pricing strategies, and skin-type suitability due to the large volume of product data available.

This project addresses these challenges by analyzing cosmetic product information, including brand, category, price, ranking, and skin-type suitability. Using Tableau, the data is transformed into interactive dashboards that enable users to explore key performance indicators (KPIs), compare products, and identify meaningful market trends to support informed business decisions.

### Customer Problem Statement Table:

| I AM | I’M TRYING TO | BUT | BECAUSE | WHICH MAKES ME FEEL |
|---|---|---|---|---|
| A cosmetic company manager, product analyst, retailer, or marketing executive. | Analyze cosmetic products, compare brands, study pricing trends, evaluate product rankings, and understand skin-type suitability through interactive dashboards. | I find it difficult to interpret large amounts of cosmetic product data and identify meaningful insights. | The data is scattered, requires cleaning, and traditional spreadsheets do not provide interactive analysis or KPI-based reporting. | Uncertain about business decisions related to product strategy, pricing, and marketing. |

## CUSTOMER PROBLEM SUMMARY

Stakeholders require an interactive and easy-to-use dashboard that transforms cosmetic product data into meaningful insights. The solution should enable users to identify high-performing brands, compare pricing strategies, analyze product categories, evaluate customer rankings, and understand product suitability across different skin types. These insights help organizations make confident, data-driven decisions that improve business performance and customer satisfaction.

## IMPACT

- **Limited Business Insights** – Difficulty identifying top-performing brands, products, and categories.
- **Inefficient Decision-Making** – Manual analysis of cosmetic data is time-consuming and prone to errors.
- **Suboptimal Marketing Strategies** – Lack of clear consumer insights affects targeted promotional campaigns.
- **Pricing Challenges** – Difficulty evaluating pricing trends across competing brands.
- **Reduced Customer Satisfaction** – Limited understanding of skin-type suitability may lead to poor product recommendations and lower customer engagement.

## Problem Statement (PS-1):

| Problem Statement | I am (customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| PS-1 | A cosmetic company manager, product analyst, retailer, or marketing executive looking to understand consumer preferences and improve business performance. | Analyze cosmetic products, compare brand performance, pricing trends, product rankings, and skin-type suitability using interactive Tableau dashboards. | I struggle with raw and unorganized cosmetic data and cannot quickly identify meaningful KPIs and market trends. | Traditional reports and spreadsheets do not provide interactive visual insights, making analysis slow and less effective. | Confused and uncertain, risking poor business decisions, ineffective marketing strategies, and reduced customer satisfaction. |

## 2.2 Empathy Map Canvas

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### Empathy Map Canvas

An empathy map helps understand the behaviour, needs, and expectations of stakeholders in the cosmetics industry. It enables cosmetic companies, retailers, and marketing teams to identify consumer preferences, evaluate product performance, and understand market trends. Before developing the dashboard, it is important to understand what users think, say, do, and feel so that the solution effectively addresses their business challenges.

### Develop Shared Understanding and Empathy

| Says | Think | Does | Feels |
|---|---|---|---|
| "Which brand has the best-performing products?" "Which products are suitable for different skin types?" "How can we improve sales and customer satisfaction?" | Wants a single dashboard with clear KPIs. Believes visual analytics can reveal hidden business opportunities. Wants accurate insights for better decision-making | Reviews product performance reports, compares brands and prices, analyzes customer preferences, and plans marketing strategies | Overwhelmed by scattered product data. Frustrated by manual analysis. Confidence when dashboards provide clear and actionable insights. |

**PAIN –** Scattered cosmetic data, time-consuming manual analysis, difficulty identifying product trends, and limited visibility into brand performance.

**GAIN –** Clear KPIs, interactive dashboards, easy comparison of brands and prices, better product planning, targeted marketing, and informed business decisions.

# 2.3 Brainstorming

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### Step-1: Team Gathering, Collaboration and Selection of the Problem Statement

The team collaborated to identify a meaningful business problem in the cosmetics industry. After discussing several ideas such as brand comparison, product recommendation, pricing analysis, and skin-type suitability, the team finalized the project "Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau." The selected problem addresses the need for understanding consumer preferences, product performance, and market trends through interactive data visualization.

## Step-2: Brainstorming, Idea Listing and Grouping

The team generated and organized ideas that would provide maximum business value. The ideas were grouped into the following categories:

- **Business KPIs:** Total Products, Total Brands, Average Price, Average Product Rank, Product Categories.
- **Brand & Product Analysis:** Brand-wise product distribution, category-wise analysis, top-ranked products.
- **Consumer Insights:** Skin-type suitability analysis, price comparison across brands, category trends.
- **Dashboard & Story:** Develop an interactive Tableau dashboard and Story to present business insights and recommendations.

## Step-3: Idea Prioritization

| Idea | Feasibility | Impact | Priority |
|---|---|---|---|
| KPI Cards (products, Brands, Average Price, Average Rank) | High | High | P1 |
| Brand-wise Product Analysis | High | High | P1 |
| Category-wise Analysis | High | High | P1 |
| Skin-Type Suitability Analysis | High | Medium | P2 |
| Interactive Dashboard & Tableau | High | High | P1 |
| AI-Based Product Recommendation | Medium | Medium | P3 (Future Scope) |

# 3. REQUIREMENT ANALYSIS

## 3.1 Customer Journey Map

The customer journey map illustrates the end-to-end experience of a cosmetic company manager, retailer, or business analyst using the Cosmetic Insights Dashboard—from data collection to actionable business insights.

| Stage | User Action | Touchpoint | Emotion |
|---|---|---|---|
| Awareness | Realizes cosmetic product data is difficult to analyze manually. | Excel / CSV | Frustrated |
| Data Collection | Downloads the cosmetics dataset from kaggle or other sources. | Kaggle / CSV | Hopeful |
| Preparation | Cleans missing values, charts, and dashboard in Tableau | Tableau Prep / Excel | Focused |
| Visualization | Create KPI cards, charts, and dashboard in Tableau | Tableau Desktop/ Tableau Public | Engaged |
| Insight | Identifies top brand, pricing trends, product categories, and skin-type suitability. | Dashboard / Story | Confident |
| Action | Uses insights for product planning, pricing, inventory, and marketing decisions. | Business Operations | Satisfied |

## 3.2 Solution Requirements

| FR No. | Functional Requirements | Sub-Requirement (Story / Task) |
|---|---|---|
| FR-1 | Data Collection | Download the Cosmetics dataset (CSV) from Kaggle or another reliable source |
| FR-2 | Data Cleaning | Handle missing values, remove duplicates, standardize brand and category names, and correct data types. |
| FR-3 | Calculated Fields | Create calculated fields such as Average Price, Average Rank, Product Count, and Brand Count. |
| FR-4 | KPI Cards | Total Product, Total Brands, Average Price, Rank, Product Categories. |
| FR-5 | Visualizations | Brand-wise analysis, category distribution, price analysis, skin-type suitability, product Rankings |
| FR-6 | Interactive Dashboard | Filters(Brand, Category, Skin Type),tooltips, and interactive dashboard action. |
| FR-7 | Tableau Story | Publish a Tableau Story presenting KPIs, market trends, insights, and business recommendations. |
| FR-8 | Publishing | Publish the Tableau workbook on Tableau Public with a shareable link. |

## Non-Functional Requirements

| NFR No. | Non-Functional Requirements | Description |
|---|---|---|
| NFR-1 | Usability | The dashboard should be user-friendly with clear labels, tooltips, and a consistent color theme. |
| NFR-2 | Performance | The dashboard should load efficiently on Tableau Public and provide smooth interaction |
| NFR-3 | Reliability | All KPIs and visualizations should accurately represent the cleaned cosmetics dataset |
| NFR-4 | Scalability | The dashboard should support additional brands, products categories, and future datasets without major redesign. |
| NFR-5 | Availability | The published Tableau Public dashboard should be publicly accessible at all time. |

# 3.3 Data Flow Diagram

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## Data Flow – Level 1 (Project Overview)

The DFD below describes how raw cosmetics data flows through cleaning, transformation, visualization, and finally to the end user through the Tableau dashboard.

| S.No | Process | Description |
|---|---|---|
| 1 | Data Source | Raw Cosmetics dataset (CSV) downloaded from kaggle. |
| 2 | Data Cleaning | Handle missing values, remove duplicates, standardize brand and category names, and correct data types using Excel/Tableau prep. |
| 3 | Data Transformation | Create calculated fields such as average price, average rank, product count, and Brand Count. |
| 4 | Data Visualization | Connect the cleaned data to Tableau and create KPI cards and visualizations. |
| 5 | Dashboard & Story | Combine charts into an interactive dashboard and Tableau Story with business insights and recommendations. |
| 6 | User Interaction | Users apply filters, explore visualizations, and derive insights from the published dashboard. |

## 3.4 Technology Stack

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### Technical Architecture

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | Data Source | Cosmetics dataset(CSV) | Kaggle / CSV |
| 2 | Data Cleaning | Handle missing values, remove duplicates, standardize data | Excel, Tableau Prep |
| 3 | Data Storage | Cleaned CSV / Tableau Extract | Local File System |
| 4 | Visualization Engine | KPI Cards, Charts, Dashboards, Tableau Story | Tableau Desktop / Tableau Public |
| 5 | Publishing | Public interactive dashboard | Tableau Public |
| 6 | Version control | Dataset, workbook, screenshots, documentation | Git & GitHub |
| 7 | Documentation | Project report and supporting documents | MS Word / PDF / Google Drive |

# 4. PROJECT DESIGN

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## 4.1 Problem–Solution Fit

| Block | Details |
| --- | --- |
| Customer Segment(s) | Cosmetic companies, product managers, retailers, business analysts, and marketing teams. |
| Jobs-to-be-Done / Problems | Understand brand performance, pricing trends, product rankings, and skin-type suitability to support business decisions. |
| Triggers | Declining product performance, changing customer preferences, ineffective pricing strategies, and unclear market trends. |
| Emotions Before / After | Before: Confused by scattered product data. After: Confident with clear KPIs and interactive dashboards. |
| Available Solutions | Excel reports, static charts, and traditional BI reports with limited interactivity |
| Customer Constraints | Limited analytics expertise, time constraints, and difficulty managing large datasets. |
| Behaviour | Users regularly review product data but struggle to identify trends quickly. |
| Channels of Behaviour | Online: Tableau Public Dashboard. Offline: Business reports and presentations. |
| Problem Root Cause | Cosmetic data is scattered, unorganized, and difficult to analyze using traditional methods. |
| Your Solution | A cleaned cosmetics dataset with an interactive Tableau dashboard and story that presents KPIs, market trends, and business insights in one place. |

## 4.2 Proposed Solution

| S.No | Parameter | Description |
|---|---|---|
| 1 | Problem Statement | Cosmetic companies and analysts cannot easily analyze raw product data to identify market trends and key business insights. |
| 2 | Idea / Solution | Develop an interactive Tableau dashboard and story using a cleaned cosmetics dataset with KPI cards and visual analytics. |
| 3 | Novelty / Uniqueness | Combines KPIs, brand analysis, pricing trends, skin-type suitability, and interactive dashboards in a single Tableau Public workbook. |
| 4 | Social Impact / Customer Satisfaction | Helps customers identify suitable cosmetic products while enabling businesses to improve product planning and marketing strategies. |
| 5 | Business Model / Revenue Model | Supports data-driven business decisions that improve product performance, pricing strategies, and customer engagement. |
| 6 | Scalability | Can be extended to include additional brands, product categories, customer reviews, and future datasets without redesigning the dashboard. |

## 4.3 Solution Architecture

| S.No | Layer | Responsibility | Technology |
|---|---|---|---|
| 1 | Source Layer | Raw cosmetics dataset ingestion | Kaggle CSV |
| 2 | Preparation Layer | Data cleaning, standardization, and calculated fields | Excel / Tableau Prep |
| 3 | Storage Layer | Cleaned CSV / Tableau Extract | Local File System |
| 4 | Visualization Layer | KPI cards, charts, dashboard, and Tableau Story | Tableau Desktop / Tableau Public |
| 5 | Presentation Layer | Public interactive dashboard and shareable link | Tableau Desktop / Tableau Public |

# 5. PROJECT PLANNING & SCHEDULING

## 5.1 Project Planning

### Product Backlog, Sprint Schedule and Estimation

| Sprint | Functional Requirement | User Story / Task | Story Points | Priority | Team Members |
|---|---|---|---|---|---|
| Sprint-1 | Data Collection | Download the Cosmetics dataset from Kaggle. | 2 | High | Gayathri |
| Sprint-1 | Data Cleaning | Handle missing values, remove duplicates, and standardize brand and category names. | 5 | High | Gayathri |
| Sprint-1 | Calculated Fields | Create Average Price, Average Rank, Product Count, and Brand Count. | 3 | High | Gayathri |
| Sprint-1 | Data import | Connect the cleaned dataset to Tableau. | 2 | Medium | Sana |
| Sprint-1 | Repository Setup | Create GitHub repository and organize project files. | 2 | Medium | Sana |
| Sprint-2 | KPI Cards | Build KPI cards for | 5 | High | Chaturya |

| | | | | | |
|---|---|---|---|---|---|
| | | Total Products, Total Brands, Average Price, and Average Rank. | | | |
| Sprint-2 | Charts | Create Brand Analysis, Category Distribution, Price Analysis, and Skin-Type charts. | 5 | High | Chaturya |
| Sprint-2 | Dashboard | Develop an interactive dashboard with filters, tooltips, and dashboard actions. | 4 | High | Revathi |
| Sprint-2 | Story & Publish | Create Tableau Story, publish the workbook, and prepare the project demonstration. | 6 | High | Revathi |

## Sprint Duration & Velocity

| Sprint | Total Story Points | Duration | Start Date | End Date | Status |
|---|---|---|---|---|---|
| Sprint-1 | 14 | 6 Days | 20 June 2026 | 25 June 2026 | completed |
| Sprint-2 | 20 | 8 Days | 26 Jun 2020 | 03 July 2026 | completed |

## Velocity Calculation

Velocity = Total Story Points Completed / Number of Sprints

Total Story Points = 14 + 20 = 34

Number of Sprints = 2

Velocity = 34 / 2 = 17 Story Points per Sprint

## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

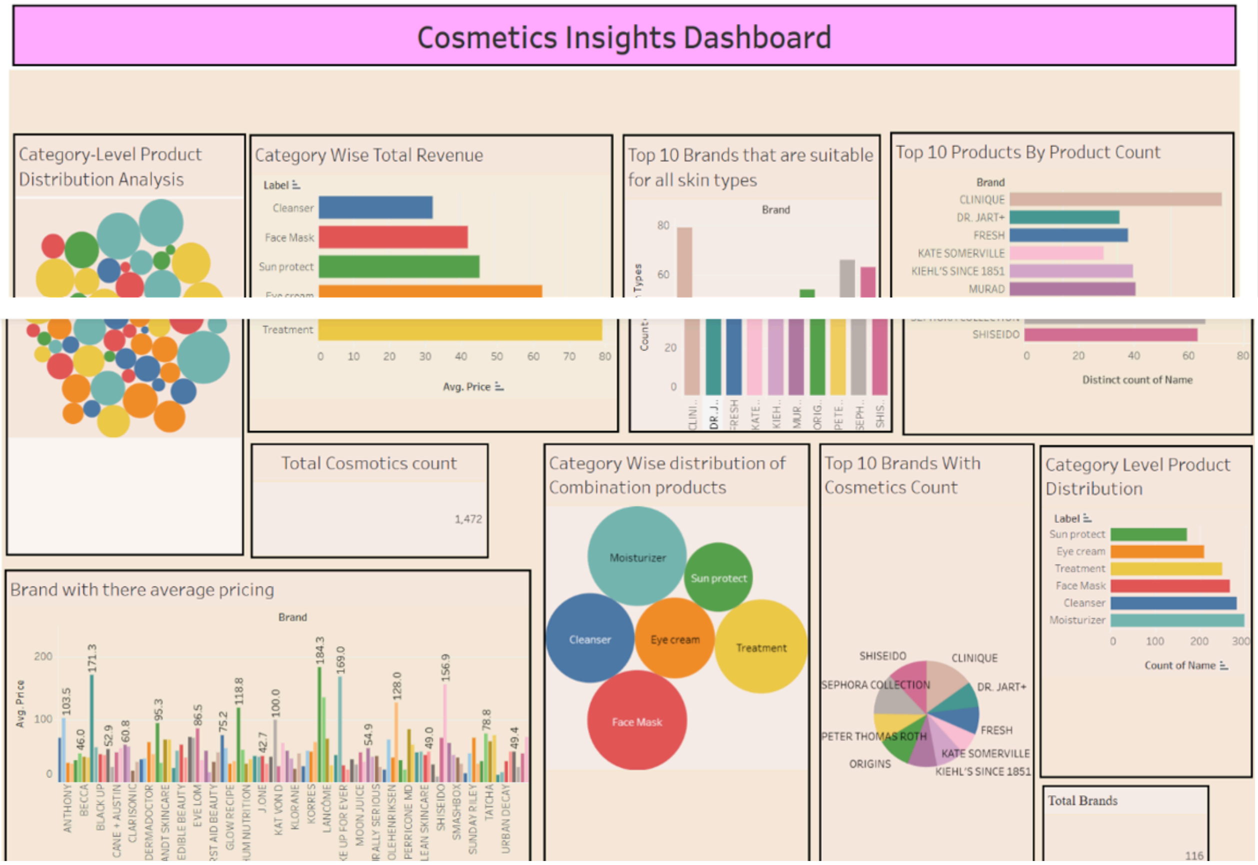
The dashboard was tested to ensure the correctness of KPIs, filter functionality, dashboard interactivity, loading performance

, and accessibility on Tableau Public.

| Test ID | Test Case | Expected Result | Actual Result | Status |
|---|---|---|---|---|
| T-01 | KPI Cards Display | Total Products, Total Brands, Average Price, Average Rank displayed correctly | All KPI cards displayed correctly | Pass |
| T-02 | Brand Filter | Charts update based on the selected brand | Charts updated correctly | Pass |
| T-03 | Category Filter | Charts update based on the selected product category | Charts updated correctly | Pass |
| T-04 | Skin Type Filter | Charts update based on the selected skin type | Charts updated correctly | Pass |
| T-05 | Tooltips | Tooltips display accurate product information and values | Tooltips displayed correctly | Pass |
| T-06 | Dashboard Load Time | Dashboard loads within 5 seconds on Tableau Public | Loaded within 3–4 seconds | Pass |
| T-07 | Story Navigation | All Tableau Story points navigate correctly | Navigation works correctly | Pass |
| T-08 | Cross-Browser | Dashboard | Successfully | Pass |

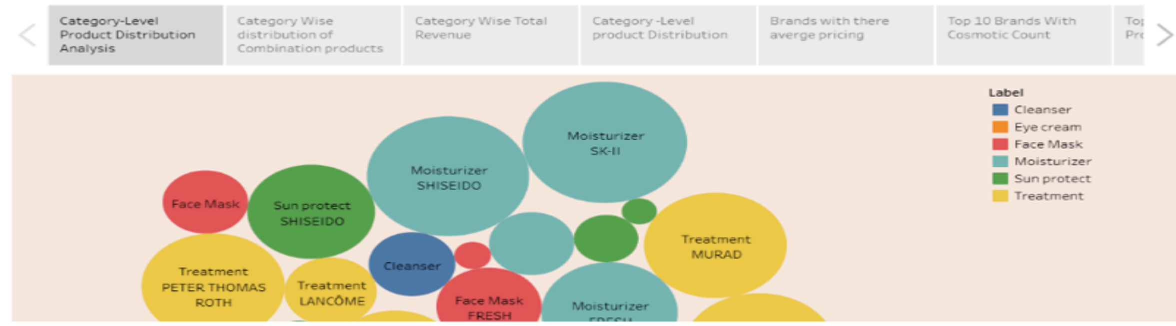
| | | | | |
|---|---|---|---|---|
| | Compatibility | works on Chrome, Edge, and Firefox | tested on all browsers | |
| T-09 | Mobile Responsiveness | Dashboard is viewable on mobile and tablet devices | Layout adjusted and displayed correctly | Pass |
| T-10 | Data Accuracy | KPI values match the cleaned cosmetics dataset | Values matched the source dataset | Pass |

**Interactive Dashboard**



**Interactive Tableau Story**

Story 1




## 8. ADVANTAGES & DISADVANTAGES

**Advantages**

Provides a single, interactive dashboard for analyzing cosmetic product trends.

Helps identify top-performing brands and product categories.

Enables comparison of product prices, ratings, and customer preferences.

Assists businesses in understanding skin type suitability across products.

Supports data-driven decision-making for marketing and product development.

Easy to share through Tableau Public without requiring software installation.

Can be expanded with new products, brands, or future datasets.

**Disadvantages**

Results depend on the accuracy and completeness of the dataset.

Tableau Public makes the workbook and data publicly accessible, making it unsuitable for confidential information.

The dashboard uses historical data and does not provide real-time updates.

Predictive analytics, such as sales or rating forecasting, is not included.

Mobile viewing experience may be less optimized than desktop.

## 9. CONCLUSION

The Cosmetic Insights project successfully transformed raw cosmetic product data into an interactive Tableau dashboard that provides meaningful business insights. The dashboard enables users to analyze product categories, brand performance, pricing trends, customer ratings, and skin type suitability through clear visualizations and KPI cards. These insights help businesses, retailers, and consumers make informed decisions based on data rather than assumptions. Overall, the project demonstrates how Tableau can effectively convert complex datasets into simple, interactive, and valuable visual analytics.

## 10. FUTURE SCOPE

Integrate live cosmetic product data through APIs for real-time analysis.

Add machine learning models to predict product ratings and customer preferences.

Perform customer review sentiment analysis using Natural Language Processing (NLP).

Expand the dataset with additional brands, regions, and product categories.

Develop role-based dashboards for business managers, marketers, and analysts.

Deploy the dashboard as a web application for easier public access.

Generate automated reports and email summaries for stakeholders.

## 11.APPENDIX

Source Code
All source files (Tableau workbook .twbx, cleaned dataset, screenshots, README) are maintainedinthe GitHub repository listed below. Dataset Link
Cosmetic Insights Dataset — available in the /data folder of the GitHub repository
GitHub & Project Demo Link
GitHub Repository:
https://github.com/GayathriGollapalli/Cosmetic-Insights-Dashboard
Tableau Public Profile:
https://public.tableau.com/app/profile/gollapalli.rishitha.gayathri/vizzes

**Team Details:**
Team ID:
Team Members:
Abdul Hafeez Shaik
Avanigadda Dola Chaturya
Gollapalli Rishitha Gayathri
Golla Sana Grace
Revathi Gorupina
Domain: Data Analytics & Business Intelligence
Tool: Tableau Public 2026.2